

Question

AgO reacts with KIO_4 under alkaline conditions, undergoing a formal "disproportionation reaction" to obtain two silver-containing substances. The anion part of one of the compounds, **A**, is a complex ion of Ag and the coordination number of Ag is 4. The mass fraction of Ag in this compound is 14.36%. Deduce the chemical formula of **A** and write the reaction equation.

There are the following statements, which of the statements are true?

1. In the anion of compound **A**, the sum of the number of atoms in all ligand molecules of Ag is 16.
2. The product of the coefficients in front of each compound in the simplest reaction equation is 192.
3. Silver in compound **A** has a +1 valence.
4. The reaction also produces another silver compound with a molecular weight greater than 250.

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **F**

Detailed Explanation

Coordination number 4 Ag is usually +2 or +3 valent. The problem mentions "disproportionation", so the coordination ion is likely to be +3 valent Ag.

CHECKPOINT

0.5 PTS

The coordination ion is +3 valent Ag

Considering the size of an Ag atom, it is unlikely that there are four large-volume ligands around it. At the same time, IO_6^{5-} is a common bidentate ligand, so it is speculated that the anion part is $\text{Ag}(\text{IO}_6)_2^{7-}$.

CHECKPOINT

1 PTS

The ion coordinated with Ag is the bidentate ligand IO_6^{5-}

According to the mass fraction of Ag, it can be calculated that $M_A = M_{\text{Ag}}/14.36\% = 751.4$. The molecular weight of the remaining part excluding silver and periodate is 197.7. The main cation in this reaction is K^+ . According to the molecular weight, it can be calculated that compound **A** contains five K^+ and two H^+ atoms. The chemical formula is written as $\text{K}_5\text{H}_2\text{Ag}(\text{IO}_6)_2$. In fact, two hydrogen atoms are combined with IO_6^{5-} , which does not affect its coordination ability.

CHECKPOINT

0.5 PTS

$$M_A = 751.4$$

CHECKPOINT

1 PTS

The chemical formula of **A** is written as $K_5H_2Ag(IO_6)_2$

In the anion of compound **A**, the ligand molecules of **Ag** are two HIO_6^{4-} , and the sum of the number of atoms is 16, so statement 1 is correct.

The valence of silver is +3, so statement 3 is wrong.

CHECKPOINT

1 PTS

The ligand of **Ag** is HIO_6^{4-}

According to the "disproportionation reaction", the other substance is +1 valent **Ag**, which is most reasonably written as Ag_2O under alkaline conditions, so the equation is



CHECKPOINT

0.5 PTS

The other product is Ag_2O

CHECKPOINT

1 PTS

The equation is



Hydrogen can also be written on periodate.

The product of the coefficients of the simplest reaction equation is 192, so statement 2 is correct. The other silver compound produced by the reaction is Ag_2O , and the molecular weight is 231.8, so statement 4 is wrong.